Area Modeling: Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$

Polynomial Equations & Factoring • Standard: CCSS.MATH.HSA.APR.A.1

BOOK STORY OPENER: From Al-Khwarizmi's Physical Squares to the 1535 Duel in Milan & Kahan's [POLY] Solver

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by literally arranging physical square and rectangular tiles to 'complete the square'—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! Students build physical $(x + a)(x + b)$ tile rectangles and verify roots with the SC-32 [POLY] solver.

Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Difference of Two Squares) & Thinking in the Stack (SC-32 Ch. 7: The Duel in Milan & [POLY])

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Algebra Area Tiles & Factoring Frame

Place 1 x^2 tile, 5 x-tiles, and 6 unit-1 squares on the factoring stage.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Arrange into a solid rectangle: width is $(x + 2)$, length is $(x + 3)$. Total Area = $x^2 + 5x + 6$.

3. CALCULATOR ACTION (SC-32 KEYCAPS & STACK TRACE)

Keystroke Flow: [1] [ENTER] [5] [ENTER] [6] [↗] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Coefficient a = 1 entered into X
[ENTER]	1	1 lifted to Y
[5]	5	Coefficient b = 5 entered into X
[ENTER]	5	b in Y, a in Z
[6]	6	Coefficient c = 6 entered into X
[↗] [POLY]	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The physical dimensions of the rectangle (x+2) and (x+3) correspond to roots x = -2 and x = -3 solved by the [POLY] quadratic pipeline.

Core Investigation Problem:

Arrange tiles for $x^2 + 7x + 12$ into a rectangle to find its factors, then verify with [POLY].

Zero Pairs: Factoring Trinomials with Negative Terms

Factoring Quadratic Trinomials • Standard: CCSS.MATH.HSA.REI.B.4

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Set up 1 x^2 tile and 6 negative unit tiles for $x^2 - x - 6$. Prediction: Add zero pair (+2x and -3x): forms a rectangle of (x - 3) by (x + 2).

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[1] [ENTER] [1] [+/−] [ENTER] [6] [+/−] [↗] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	Coefficient a = 1
[ENTER]	1	1 lifted to Y
[1]	1	Enter magnitude of b
[+/−]	-1	Coefficient b = -1 in X
[ENTER]	-1	b in Y, a in Z
[6]	6	Enter magnitude of c
[+/−]	-6	Coefficient c = -6 in X
[↗] [POLY]	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: x - 3), Root 2 = -2 in X (factor: x + 2)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding (x + 2)(x + 3) as x^2 + 6 by forgetting the two linear cross-terms (2x and 3x).

Hands-On Block Remediation: Build the rectangle on the Algebra Factoring Stage with 1 [x^2] tile, 5 [x] strips, and 6 [1] unit tiles so the four area quadrants (x^2, 3x, 2x, 6) are physically visible.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block Remediation: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

4. EXTENSION MATHEMATICAL INSIGHT

Zero-pair tiles cancel to zero while allowing rectangular dimensions to complete. [POLY] computes the real roots directly.

Tactile Algebra: Area Models, Polynomials & Quadratic Factoring

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: AREA MODELING & QUADRATIC FACTORING RULES

- Polynomial tiles represent algebraic terms: large square = x^2, rectangle = x, small square = 1.
- An algebraic expression factors when tiles form a solid rectangle with dimensions (x + a) by (x + b).
- Zero pairs: positive (+x) and negative (-x) tiles cancel out to zero to complete rectangular arrays.

Key Sequence: [1] [ENTER] [5] [ENTER] [6] [↷] [POLY]

Problem 1: Factor $x^2 + 6x + 8$ into a rectangle of algebra tiles, then verify with [POLY].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 2: Factor $x^2 - 4$ (difference of two squares) using zero-pair x-tiles.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 3: Factor $x^2 + 5x + 4$ using polynomial tiles and verify.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 4: Factor $x^2 - 5x + 6$ using zero pairs.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Unit 08: Tactile Algebra: Area Models, Polynomials & Quadratic Factoring — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 7–8

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Arrange tiles for $x^2 + 7x + 12$ into a rectangle to find its factors, then verify with [POLY].

Core Solution: (x + 3)(x + 4), roots -3 and -4. Keystrokes: [1] [ENTER] [7] [ENTER] [1][2] [g:][POLY]

Core Check Keys: 1 ENTER 7 ENTER 1 2 ↷ POLY

Extension Challenge: Factor $x^2 - 5x + 6$ using zero pairs on the Algebra Stage and verify on the calculator.

Extension Solution: (x - 2)(x - 3), roots 2 and 3. Keystrokes: [1] [ENTER] [5] [+/-] [ENTER] [6] [g:][POLY]

Ext. Check Keys: 1 ENTER 5 +/- ENTER 6 ↷ POLY

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Factor $x^2 + 6x + 8$ into a rectangle of algebra tiles, then verify with [POLY].

Expected Answer: (x + 2)(x + 4); roots -2, -4

SC-32 Verification Keys: 1 ENTER 6 ENTER 8 ↷ POLY

Problem 2: Factor $x^2 - 4$ (difference of two squares) using zero-pair x-tiles.

Expected Answer: (x - 2)(x + 2); roots 2, -2

SC-32 Verification Keys: 1 ENTER 0 ENTER 4 +/- ↷ POLY

Problem 3: Factor $x^2 + 5x + 4$ using polynomial tiles and verify.

Expected Answer: (x + 1)(x + 4); roots -1, -4

SC-32 Verification Keys: 1 ENTER 5 ENTER 4 ↷ POLY

Problem 4: Factor $x^2 - 5x + 6$ using zero pairs.

Expected Answer: (x - 2)(x - 3); roots 2, 3

SC-32 Verification Keys: 1 ENTER 5 +/- ENTER 6 ↷ POLY

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.